

CO5_AT3_SIMATS Compiler Level3 Exam

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QUESTIONS

LEVEL 3 — HARD

Q1

Numbers

Q2

Control

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Level 3 — Hard (H) Steps — Control

Given an integer num, return the number of steps to reduce it to zero. In one step, if the current number is even, you have to divide it by 2, otherwise, you have to subtract 1 from it.

Example 1:

Input: num = 14

Output: 6

Explanation:

Step 1) 14 is even; divide by 2 and obtain 7.

Step 2) 7 is odd; subtract 1 and obtain 6.

Step 3) 6 is even; divide by 2 and obtain 3.

Step 4) 3 is odd; subtract 1 and obtain 2.

Step 5) 2 is even; divide by 2 and obtain 1.

Step 6) 1 is odd; subtract 1 and obtain 0.

Testcase:1. n = 5

2. n = 10

3. n = 12

4. n = 18

5. n = 20

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         int num = sc.nextInt();
7         int steps = 0;
8         while (num != 0) {
9             if (num % 2 == 0) {
10                 num = num / 2;
11             } else {
12                 num = num - 1;
13             }
14             steps++;
15         }
16         System.out.println(steps);
17     }
18 }
```

Program Input (stdin)

14

Output

6

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